FULL CARD 29 — Probability, Expected Frequency and Finance

MATHEMATICS • PROBABILITY • PERCENTAGES • FINANCIAL LITERACY

PROBABILITY • For equally likely outcomes, P(event) = favourable outcomes ÷ total outcomes. Probabilities range from 0 to 1.

A bag contains 4 red, 3 blue and 1 green counter. One counter is selected at random.

Event	Probability	Decimal / percentage
Red	4/8 = 1/2	0.5 = 50%
Blue	3/8	0.375 = 37.5%
Green	1/8	0.125 = 12.5%
Not green	7/8	0.875 = 87.5%

Expected frequency

If the experiment is repeated 240 times, the expected number of green outcomes is 240 × 1/8 = 30. This is a long-run expectation, not a guarantee of exactly 30 green results.

Financial mathematics

Method	Formula / calculation	Result
Simple interest	I = Prt; ■5,000 × 0.04 × 3	■600 interest
Compound interest	A = P(1 + r)■; ■5,000(1.04)³	■5,624.32 approx.
Discount	20% of ■1,500 = 0.20 × 1,500	■300 off; pay ■1,200
Unit price	■180 for 3 kg = 180 ÷ 3	■60 per kg

Comparison: Simple interest is calculated on the original principal each period. Compound interest is calculated on principal plus accumulated interest. Always check whether fees, tax, inflation, or changing rates are excluded.

Data literacy: A sample may be biased if some groups are less likely to be selected or respond. Larger samples can reduce random variation but do not automatically remove bias.

Practice: A fair six-sided die is rolled. Find P(number greater than 4). Then calculate the sale price of a ■2,400 item after a 12.5% discount.